Raj Jha

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Data Scientist with over four years of experience specializing in machine learning, deep learning, Gen AI, LLMs, and natural language processing. Proven track record in developing scalable models and algorithms that drive business improvements and innovations at major tech companies like Wayfair, Swiggy, Eli Lilly. Proficient in Python, SQL, and PySpark, with strong analytical skills and a solid academic background from the IISc Bengaluru.

Education

Date	Degree	Institute	Performance
2020-2022	M.Tech - Computer Science & Engineering, specialization in Artificial Intelligence	Indian Institute of Science Bangalore	**8.3/**10
2015-2019	B. Tech - Computer Science & Engineering	Silicon Institute of Technology Bhubaneswar	**9.34**/10

Professional Experience

- **Machine Learning Scientist 2, Wayfair (Bengaluru)** (*June 2024 - Present*)
  - **Semantic Product Search for Wayfair Product Catalog**
    - Designed and implemented a semantic product search system using **Sentence-BERT (s-BERT)** to enhance product discovery and improve search relevance.
    - Engineered a **two-tower architecture** model for semantic similarity to encode customer queries and product descriptions.
    - Achieved significant improvements in search relevance with an **NDCG@10** uplift of 24% in offline experiments and increased engagement metrics in live A/B tests.
  - **Autosuggest Ranking**
    - Developed a **linear ranker model** to improve the ranking of query suggestions in the autosuggest feature, optimizing for relevance, engagement, and conversion.
    - Utilized **logistic regression and the LambdaMART model** to rank suggestions by their likelihood of driving user engagement and conversions.
    - Deployed the ranker model into the **ElasticSearch pipeline**, ensuring seamless integration with existing autosuggest workflows. Enabled real-time ranking of suggestions for millions of user queries with low latency.
    - Observed a 15% uplift in **CTR** and an 8% improvement in **order conversion rate** during live A/B testing, alongside consistent performance gains in offline metrics like **NDCG@5** and **MRR**.

- **Data Scientist, Swiggy (Bengaluru)** (*May 2022 - June 2024*)
  - **Real-time Restaurant Recommendation System**
    - Developed a real-time restaurant recommendation system, utilizing proximity-based exponential deboosting and demand shaping to provide personalized restaurant suggestions that adapt to user preferences and geographical data.
    - This resulted in a 31p gain per order in UE at a platform level in **India**, significantly enhancing the user experience by delivering personalized recommendations. **NDCG@k and MRR** are also improved by 0.2pp in offline evaluation.

  - **Review summarization on the Menu Page**
    - Implemented review summarization using Gen AI technology hosted on **Google Cloud AI Platform** integrated with **BigQuery** and **Vertex AI** for training, leveraging natural language processing techniques to analyze and summarize customer reviews to assist in decision-making and enhance the trustworthiness of restaurants.
    - **Achieved a 0.13% increase in M2O conversions**, reduced the IGCC (Inferred Good Customer Count) by Rs 0.44 per platform order, and lowered cancellation rates by 0.09 percentage points in the A/B experiment across India1 cities. Post-model scale-up led to an increase in daily orders by 4,000 iOPD in India1.

  - **Customer-Restaurant Embeddings:**
    - Engineered customer-restaurant embeddings with a **two-tower architecture** to deepen the personalization of offers and recommendations by capturing nuanced customer behaviors and preferences about restaurant features.
    - Contributed to a substantial increase in platform orders, adding 3.6k iOPD, thereby boosting customer satisfaction and retention.

  - **AI-Powered Customer Support:**
    - Developed an AI-powered customer support system using GPT-4 and Langchain, enabling real-time, context-aware responses to customer inquiries on an e-commerce platform.

- The architecture was designed to handle high volumes of queries with minimal latency, providing quick and accurate resolutions.
- This solution reduced response time by 30% and increased customer satisfaction by 20%, significantly improving operational efficiency and user experience.

- **Data Scientist Intern, Eli Lilly & Company (Bengaluru)** (*Jun 2021 - Aug 2021*)
  - Generated Dissimilar sentences and dissimilarity scores between the various policy documents using NLP.
  - Used **TF-IDF, Word2Vec, GLOVE, and Fine-tuned BERT** model to generate sequence embedding and computed a similarity score using Cosine and Jaccard distance.
  - Leveraged **BigQuery** and **Dataproc** for large-scale data processing and distributed computations to efficiently handle large policy document datasets.
  - **Impact**: Helps **Ethics and Compliance** affiliate partner to avoid labor-intensive and inconsistent human evaluation

- **Data Scientist, Aptus Data Labs (Bengaluru)** (*Jan 2019 - Jun 2020*)
  - **Demand Forecasting and Demand Sensing for Inventory Optimization:**
    - Developed a solution for optimal Demand Forecasting using ARIMA, LSTMs, and Ensemble Models.
    - Forecast robust to real-world events like weather changes, customer behavior change
    - Impact: Helped the client to have the optimal number of inventories in stock and save **$100K/year.**
  - **Credit Fault Risk Prediction:**
    - Build a Risk Predictor model that predicts whether a customer would be a loan payer or defaulter.
    - Implemented supervised techniques like **Logistic regression, SVM, and XGBClassfier**.
    - Best Model: XGBClassfier with **ROC AUC-score of 0.82**

Publications

- **Publications: Extractive Question Answering Using Transformer-Based LM (ICONIP-2022) (springer-chapter10)**

Skills:

Languages	Python, SQL, PySpark, C, C++
Deep Learning	CNN, RNN, LSTMs, Encoder-Decoder, Transformers, BERT, Attention Mechanisms, GAN, Generative AI, Large Language Models (LLMs), LangChain
Machine Learning	Supervised and Unsupervised ML Algorithms, SVMs, Logistic and Linear Regression, Bagging and Boosting, GBT, Clustering Algo, GMM, Decision Tree
Libraries/Framework	Scikit-Learn, NLTK, spaCy, TensorFlow, Keras, Pandas, Matplotlib, Pytorch, AWS Sage maker, Linux, Spark ML for ML pipelines, Spark Data Frames, Spark SQL, Docker, Kubernetes, MLFlow, Hive, Snowflake, Azure Databricks
Tools	GCP AI Platform, Dataproc, Vertex AI, BigQuery, Cloud Storage

Awards and Volunteer Experience

- **Most Valuable Player, Swiggy**: Recognized for contribution in developing an LM deboosting-based real-time restaurant recommendation system using Gradient Boosted Decision Trees (GBDT). The solution directly contributed to an incremental gain of 5,000 orders per day (IOPD) and a 0.31 Rs UE gain Per order.
- Awarded **"Arjun Bhuyan Best Graduate"** Gold Medal for securing the highest CGPA in BTech among all disciplines. (**University Rank – 1**)
- Secured 148th position amongst 10K+ registered Amazonhine Learning Challenge 2021 participants.
- Placement Coordinator Head at the Indian Institute of Science Bengaluru (2020-22)